J Appl Physiol 93: 1280–1286, 2002;
10.1152/japplphysiol.01270.2001.

Break point of serum creatine kinase release after endurance exercise

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Received 21 December 2001; accepted in final form 10 June 2002

Totsuka, Manabu, Shigeyuki Nakaji, Katsuhiko Suzuki, Kazuo Sugawara, and Koki Sato. Break point of serum creatine kinase release after endurance exercise. *J Appl Physiol* 93: 1280–1286, 2002; 10.1152/japplphysiol.01270.2001.—We investigated whether there is a break point of creatine kinase (CK) release after daily endurance exercise and whether CK response depends on individual physical characteristics. Fifteen healthy young men performed 90 min of bicycle exercise for 3 consecutive days. Body composition, properties of the quadriceps femoris muscle (QFM), and aerobic and anaerobic capacities were estimated before the test. Blood samples were obtained 22 times during the experimental period. Endurance exercise significantly elevated serum CK from 3 h after the first exercise session ($P < 0.05$) and gradually increased thereafter. Subjects were classified into two groups according to their peak CK values: high responders (HR; >500 IU/l of CK) and low responders (LR; <300 IU/l of CK). Peak CK values during the experimental period correlated ($P < 0.01$) with workload/cross-sectional area of the QFM ($r = 0.658$), workload/volume of the QFM ($r = 0.648$), and knee extensor strength/body mass ($r = -0.634$); however, the HR and LR groups were separated in each variable. Thus the break point of CK release after endurance exercise under these conditions is 300–500 IU/l, two or three times higher than in the resting condition, and is associated with properties of the QFM.

overtraining; muscle property; physical characteristics

CREATINE KINASE (CK) is found predominantly in muscle and is released into the circulation during muscular lesions. Therefore, serum CK activity has been theoretically expected to be useful as a marker in exercise physiology and sports medicine for the detection of muscle injury and overwork (4, 10, 22, 23, 27). However, previous studies on CK release have not clearly demonstrated its value as a marker for these states (15, 16).

Numerous studies have evaluated changes in CK activity after exercise and found that it differs markedly according to exercise conditions. For example, in isometric muscle contraction exercise, peak serum CK activity is observed relatively early, 24–48 h after exercise (3, 6, 12, 13), whereas it is seen 3–7 days after exercise in eccentric muscle contraction exercise (16, 18, 19, 24), and a biphasic pattern is observed in weight training (26). These studies used short-duration and high-intensity relative workloads to limited muscle, which damaged muscle tissue and induced CK release, so their relevance to the actual process of CK release during and after endurance exercise is not clear. There is a clear social trend toward increased physical activity, and regular physical activities with low or moderate intensity have been recommended for improving general health. However, nonathletes often experience fatigue and injury from daily exercise. This may be due to the lack of an objective marker for overtraining. Serum CK activity may act as a marker for fatigue or overwork in nonathletes, and it is therefore important to examine the effects of daily repeated aerobic exercise on serum CK activity in such subjects during and after noninjurious endurance exercise.

It is also possible that the CK response to exercise depends on the individual's physical characteristics or training background (9, 14, 15, 20, 21). Therefore, detailed studies are required on the association between serum CK activity after exercise and the body composition and other characteristics of subjects, as well as on the exercise conditions.

In a previous study, Newham et al. (17) found high and low responders after a stepping exercise. We hypothesized that there might be a break point of CK release during or after the same absolute workload exercise, depending on the individual's physical characteristics. We examined the association between serum CK activity after endurance exercise and physical characteristics such as body composition and muscle properties.

METHODS

Subjects. The subjects were 15 healthy young men (mean age 19.9 yr, range 19–21 yr) who were not involved in any physical training for athletics. This experiment was performed with the approval of the Ethical Committee of Hiro-

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saki University School of Medicine, and written, informed consent was obtained from all subjects after the purpose, content, and possible risks of the study had been explained. Subjects were required to avoid strenuous exercise before, during, and 7 days after the third exercise session, except for the experimental exercise.

Exercise protocol. All subjects were recruited to participate in 3 consecutive days of an endurance exercise program to allow observation of the break point of serum CK release after daily endurance exercise. The exercise regimen consisted of 90 min of bicycling at a set absolute workload (1.5 kp, at 60 rpm) by using a cycle ergometer (Jonas Oglaend, Sandnes, Norway) at the same time of day (between 6:00 PM and 9:00 PM) on 3 consecutive days. During the endurance exercise lasting for 90 min, the heart rate was monitored with a portable heart rate telemetry device (Sport Tester PE-3000, Polar Electro, Kempele, Finland).

Blood collection and analysis. Blood was collected via the median antebrachial vein on 22 occasions: before, 30 and 60 min after the initiation of exercise, and immediately, 1 h, 3 h, and 12 h after the completion of exercise for the first exercise session; before, and immediately, 1 h, and 12 h after exercise for the second and third exercise sessions; and daily after the third exercise session for 7 days. Blood samples were allowed to clot at room temperature for 30 min and centrifuged at 3,000 *g* for 10 min. Serum was collected and frozen at −80°C for later analysis. Serum CK activity was measured at 37°C by using a commercially available assay kit [Paramax Creatine Kinase Reagent, catalog no. B6105-4 (CK), Baxter Diagnostics, Deerfield, IL] in a kinetic enzyme analyzer. CK isozyme CK-MB was determined by using an ultraviolet method in an automatic biochemical analyzer (Du Pont Aca, E. I. du Pont de Nemours, Wilmington, DE). Free fatty acid (FFA) concentration was determined by enzymatic assay by using a test kit (Determiner NEFA755, Kyowa Medics, Tokyo, Japan).

Body composition. Before the daily endurance exercise test, all subjects were evaluated for body composition. Body mass in air (BM) was measured by using an electronic scale to the nearest 0.01 kg, with subjects wearing the same attire as when being weighed underwater. Body density was determined by using underwater weighing. Body mass under water was measured by using a load cell (AD1205-k100, A & D, Tokyo, Japan) and weighing indicator (AD4323B, A & D) to the nearest 0.01 kg. Residual lung volume was measured by using a closed-circuit, oxygen-rebreathing, nitrogen-dilution method. The percentage of body fat (%fat) and fat-free mass (FFM) were calculated from body density by using the Brozek equation (2).

*Cross-sectional area and volume of the quadriceps femoris muscle.*The cross-sectional area of the quadriceps femoris muscle (CSA-QFM) and the volume of the quadriceps femoris muscle (V-QFM) were determined by using magnetic resonance imaging (Magnetom Impact, Siemens, Erlangen, Germany).

The magnetic resonance images were obtained by using a T1-weighted, spin-echo pulse sequence (repetition time = 600 ms; echo time = 15 ms), and one excitation was used to acquire the phantom data with a 256 × 256 image matrix and a 400-mm field of view. For the leg, the upper end of the greater trochanter was landmarked, and 10-mm-thick images (3-mm spacing) were obtained over the superior pole of the patella (11).

The right CSA-QFM was determined from images by using a digitizer (Software Cosmozon, Nikon, Tokyo, Japan). CSA-QFM was considered as the maximum CSA-QFM from the upper end of the greater trochanter to the superior pole of the patella in each subject. V-QFM was estimated by multiplying the sum of the muscle cross-sectional area on each image by the distance between sections (3 mm) and the slice thickness (10 mm).

Physical performance. Isometric knee extensor strength (KES) was determined as a parameter of muscle strength, by using a measuring device (Takei, Tokyo, Japan), with the knee joint angle being ~100°.

To evaluate anaerobic work capacity, maximum anaerobic leg power (high power) was measured by pedaling at full speed by using a bicycle ergometer (Power max V, Combi, Tokyo, Japan). The initial load was fixed on the basis of the subject's body weight, and 10 s of pedaling were performed three times at 120-s intervals. Loads for the second and third pedaling trials were increased on the basis of rotation rates for the first and second pedaling trials, and the maximum anaerobic work capacity determined from 10-s full-speed pedaling three times by using three different loads (4.0–10.0 kp).

On a day more than 4 wk before measurement of isometric knee extensor strength and maximum anaerobic power, each subject underwent a test of maximal oxygen consumption ($\dot{V}O_{2\,max}$) by a protocol of continuous stepwise increase in workload until exhaustion by using a bicycle ergometer. The expired air was continuously collected in a Douglas bag (Fukuda, Tokyo, Japan), and the volume of expired air was measured with a gas volume meter (Shinagawa, Tokyo, Japan). Gas analysis was performed by using oxygen and carbon dioxide analyzers (Respina IH26, NEC San-ei, Tokyo, Japan).

Statistical analysis. Data are presented as means ± SE. Differences in each parameter between the two groups were analyzed by using Mann-Whitney's nonparametric *U*-test. Changes in biochemical indexes in the two groups were analyzed by using two-way ANOVA with repeated measurements. Relationships between variables were evaluated by using simple linear regression analyses and Pearson's correlation coefficient. Probability values of $P < 0.05$ were considered significant.

RESULTS

Physical characteristics of the subjects. The physical characteristics of all subjects are shown in Table 1. %Fat and FFM were 16.7 ± 1.5% and 50.94 ± 1.21 kg, and CSA-QFM and V-QFM were 76.1 ± 3.2 cm^2 and 1,701 ± 83 cm^3, respectively. $\dot{V}O_{2\,max}$ and $\dot{V}O_{2\,max}$/BM

Table 1. *Physical characteristics of the subjects*

Variable	
Body composition	
Body mass, kg	61.8 ± 1.6
Fat-free mass, kg	50.9 ± 1.2
%Body fat, %	16.7 ± 1.5
Quadriceps femoris	
Cross-sectional area, cm^2	76.1 ± 3.2
Volume, cm^3	1,701 ± 83
Aerobic capacity	
Oxygen uptake, l/min	3.01 ± 0.12
Oxygen uptake/body mass, $ml \cdot kg^{-1} \cdot min^{-1}$	49.5 ± 2.0
Exercise intensity, %$\dot{V}O_{2\,max}$	52.2 ± 2.5
Anaerobic capacity	
Knee extensor strength, kg	66.4 ± 3.7
Leg power, W	929 ± 51

Values are means ± SE for 15 subjects. $\dot{V}O_{2\,max}$, maximal oxygen uptake.

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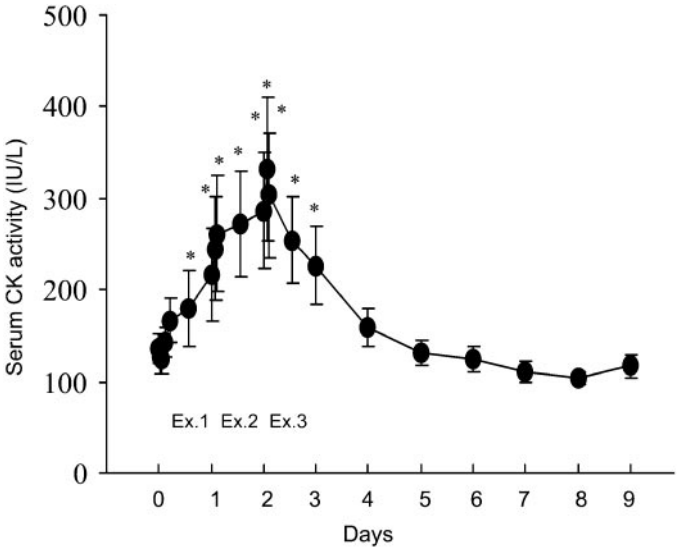


Fig. 1. Changes in serum creatine kinase (CK) activity during experimental period. Values are means ± SE. Ex 1, Ex 2, Ex 3: first, second, and third exercise session, respectively. Significant difference from preexercise value of Ex 1, *$P < 0.05$.

were 3.01 ± 0.12 l/min and 49.5 ± 2.0 ml·kg^{-1}·per min^{-1}, respectively. KES and leg power were 66.4 ± 3.7 kg and 929 ± 51 W, respectively.

Serum CK activity during experimentation. Changes in serum CK activities during the experimental periods are shown in Fig. 1. Serum CK was elevated significantly ($P < 0.05$) 3 h after the first exercise session, and it gradually increased until the third exercise session. Peak serum CK activity was observed immediately after the third exercise session (331 ± 78 IU/l). Thereafter, CK gradually decreased until *day 7*.

Changes in serum CK of classified subgroups. The subjects were classified into two groups according to the value of their peak serum CK activity after exercise: high responders (HR; >500 IU/l; *subjects A–F* in Table 2, $n = 6$) and low responders (LR; <300 IU/l; *subjects G–O*, $n = 9$). Peak CK concentrations were 751 ± 81 IU/l in HR and 184 ± 16 IU/l in LR, and these data were significantly different ($P < 0.01$) from each other (Table 3). HR individuals showed significant changes throughout the experiment, but the change in the LR group was marginal (Fig. 2). In the HR individuals, serum CK activity began to increase immediately after the first exercise session, being significantly higher ($P < 0.01$) than in the LR individuals immediately before initiation of the second exercise session (22 h after the first exercise session), and reached a peak immediately after the third exercise session, and decreased thereafter.

Table 2. *Peak CK activities during experimental period*

Subject	Peak CK Activity, IU/l
A	1,112
B	775
C	722
D	694
E	689
F	514
G	251
H	244
I	208
J	196
K	185
L	178
M	166
N	128
O	102
Mean ± SE	410 ± 81

Subjects were classified according to peak creatine kinase (CK) activity into high responders (*A–F*) and low responders (*G–O*).

Table 3. *Serum CK activities of high responders and low responders*

Serum CK Activity	High Responders ($n = 6$, CK >500 IU/l)	Low Responders ($n = 9$; CK <300 IU/l)	P Value
Peak CK, IU/l	751 ± 81	184 ± 16	<0.01*
Total CK, IU/l	6,415 ± 1,003	2,649 ± 197	<0.01*

Values are means ± SE; n, no. of subjects. *Significant difference between high responders and low responders, $P < 0.01$.

Serum CK isozyme and FFA levels. There were no significant changes in CK-MB during the experimental period in both subgroups, although these values were higher in HR than in LR between *day 2* and *day 3* (not significant) (Fig. 3).

The changes in FFA during the experimental period were not similar to those in CK activity levels. Significant changes in FFA values throughout the experiment were observed in both subgroups. In HR individuals, FFA levels tended to be lower than in the LR group, especially immediately after the third exercise session ($P < 0.05$) (Fig. 3).

Serum CK activity and physical characteristics. No differences were observed in height, BM, %fat, or FFM

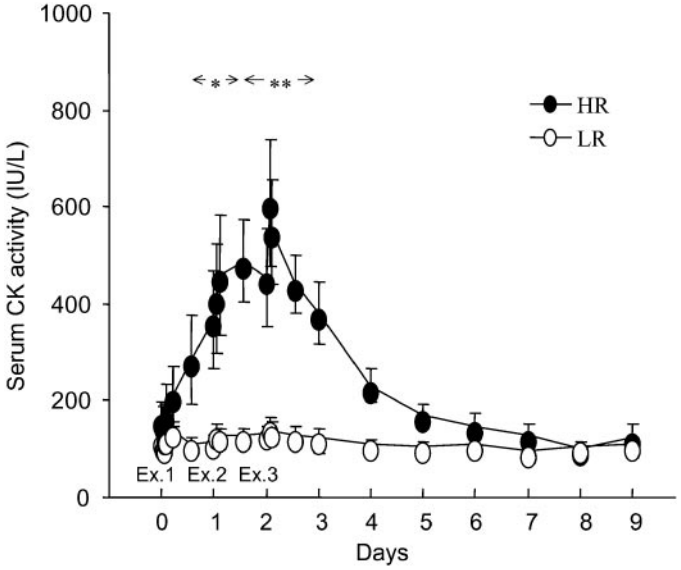


Fig. 2. Changes in serum CK activity in high responders (HR; $n = 6$) and low responders (LR; $n = 9$). Values are means ± SE. Significant difference between HR and LR: *$P < 0.05$, **$P < 0.01$.

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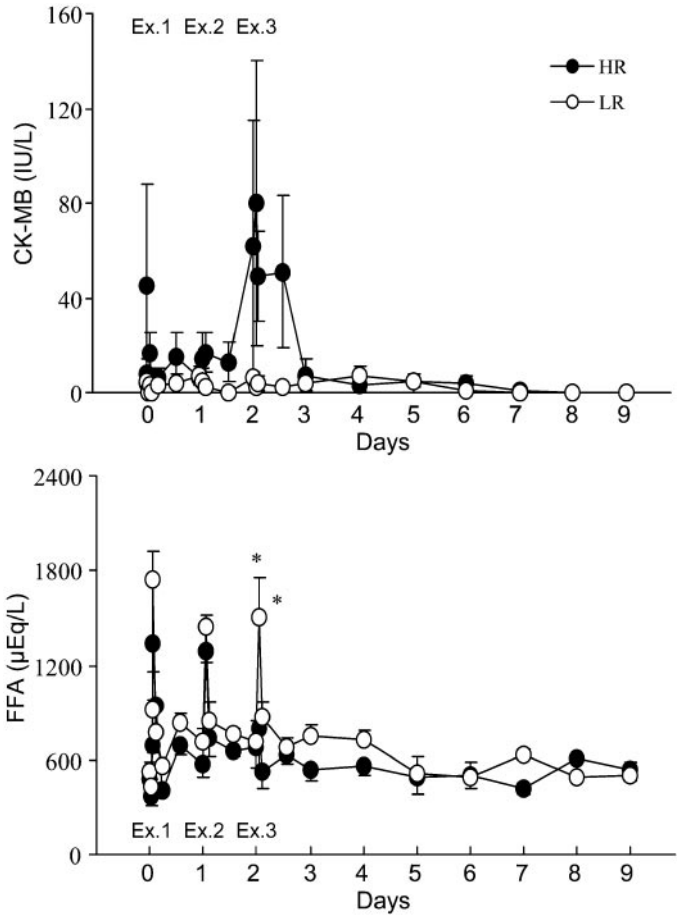


Fig. 3. Changes in CK isozyme B (CK-MB; *top*) and free fattty acid (FFA; *bottom*) concentrations during experimental period. Values are means ± SE. *Significant difference between HR and LR, $P < 0.05$.

between the HR and LR groups. However, CSA-QFM and V-QFM were lower in the HR group ($P < 0.05$ for each; Table 4). The relationships between peak CK values and leg muscle parameters in all subjects are shown in Fig. 4. Peak CK values during the experimental period correlated ($P < 0.01$) positively with the workload/CSA-QFM ($r = 0.658$) and the workload/V-QFM ($r = 0.648$). However, the HR and LR groups were separated in each variable. Although $\dot{V}O_{2\,max}$ and $\dot{V}O_{2\,max}$/BM did not significantly differ between the HR and LR groups, KES and leg power in the HR group were significantly lower ($P < 0.05$ and $P < 0.01$, respectively) than in the LR group (Table 4). The relationships between peak CK values and leg strength parameters in all subjects are shown in Fig. 5. The peak CK values during the experimental period correlated negatively with KES/BM ($r = -0.634$, $P < 0.01$). On the other hand, leg power/BM did not correlate with peak CK. The HR and LR groups were separated in each variable.

DISCUSSION

We measured serum CK at many time points in the experimental period to gain an accurate picture of changes during and after endurance exercise. The serum CK value was the balance between the value produced by exercise loading and the lost value through renal clearance.

We hypothesized that there might be a break point of CK release after the same absolute workload exercise, depending on individual physical characteristics. Although variation in each subject after exercise may be due to differences in the degree of physical workload in performing exercise, individual muscle strength is an important factor affecting serum CK activity.

A previous study (19) found no relationship between the magnitude of serum CK elevation after eccentric exercise and upper limb circumference. In contrast, we observed close associations between muscle properties and the peak serum CK level after endurance bicycle exercise. CSA-QFM, V-QFM, KES, and leg power in HR individuals were significantly lower than in the LR group, although no differences in $\dot{V}O_{2\,max}$ and $\dot{V}O_{2\,max}$/BM (which are the indexes of aerobic capacity) were observed. Furthermore, the peak CK values during the experimental period correlated positively ($P < 0.01$) with workload/CSA-QFM, workload/V-QFM, and KES/BM. However, the HR and LR groups were separated

Table 4. *Physical characteristics of high responders and low responders*

Variable	High Responders (n = 6, CK >500 IU/l)		Low Responders (n = 9, CK <300 IU/l)		P Value
Body composition					
Body mass, kg	56.6	1.7	65.2	4.9	NS
Fat free mass, kg	47.7	1.5	53.1	2.6	NS
%Body fat, %	15.6	1.7	17.4	2.3	NS
Quadriceps femoris					
Cross-sectional area, cm^2	67.3	2.6	82.1	4.0	<0.05*
Volume, cm^2	1475	70	1,852	105	<0.05*
Aerobic capacity					
Oxygen uptake, l/min	2.9	0.2	3.1	0.2	NS
Oxygen uptake/body mass, $ml \cdot kg^{-1} \cdot min^{-1}$	51.4	3.1	48.2	2.6	NS
Exercise intensity, $\%\dot{V}O_{2\,max}$	52.9	3.6	51.7	3.6	NS
Anaerobic capacity					
Knee extensor strength, kg	54.8	2.6	74.1	4.2	<0.05*
Leg power, W	816	27	1,004	74	<0.01†

Values are means ± SE; n, no. of subjects. Significant difference between high responders and low responders: *$P < 0.05$; †$P < 0.01$.

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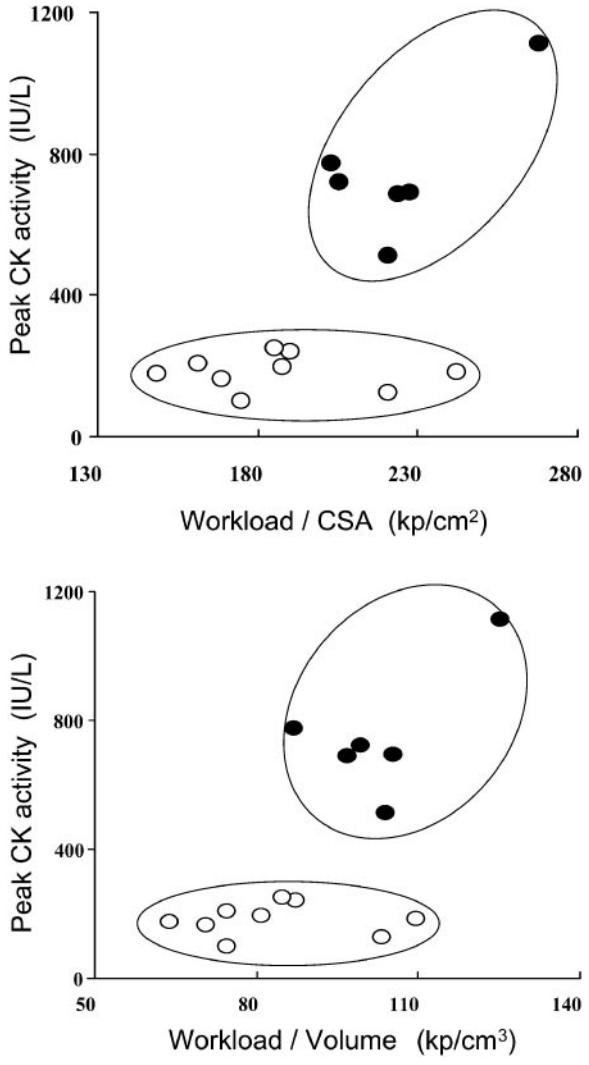


Fig. 4. Distribution of workload/cross-sectional area (CSA; *top*) and workload/volume of quadriceps (*bottom*) in HR (●) and LR (○) to peak CK values.

in each variable. This observation suggests the following mechanism. When the subjects work at a fixed loading, CK is released from the muscle when the loading exceeds a certain limit of muscle ability, and this CK value was 300–500 IU/l (CK break point). Newham et al. (17) also found that subjects could be classified into high-responder and low-responder groups after the same absolute workload stepping exercise. The peak values of high responders were below 3,000 IU/l, and those of low responders were below 400 IU/l. We estimate that the CK break point of that stepping exercise would have been ~400 IU/l, and thus it is similar to our break point value.

Although numerous studies (3, 6, 17, 19, 25) have examined the effects of exercise on serum CK activity under various exercise conditions, few studies address the break point in exercise-induced CK release. Volfinger et al. (28) indicated the possibility of a threshold value for CK release, with a "distance" threshold for CK release in horses. In the present study, when the distance was fixed experimentally, CK release was accentuated as a result of greater overload on muscle in the HR individuals, who showed lower muscle mass and strength compared with the LR group. The exercise mode and the pattern of CK response in the present study are considered to differ from those after eccentric exercise, which causes additional CK release from damaged muscle tissue.

Commonly accepted mechanisms of CK release are damage to muscle tissue or changes in myocyte membrane permeability. The pattern of CK response in the present study is considered to differ from the response after eccentric exercise, which causes additional CK release from damaged muscle tissue. With regard to membrane permeability, there are various theories of ion-distribution change, enzyme deficiency, and ATP depletion (8, 22, 29). If it is assumed that a threshold is present, the following hypothesis may be postulated. During exercise, the muscle repeatedly contracts and uses energy substrates. When the exercise intensity is within the normal range of metabolism, the muscle tissue is exercised without marked changes in membrane permeability. However, when the exercise inten-

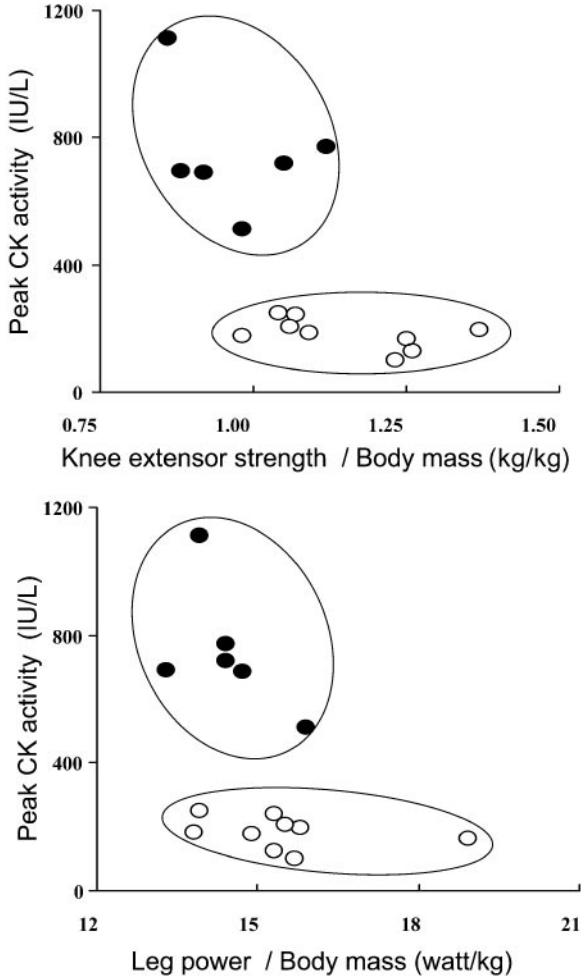


Fig. 5. Distribution of knee extensor strength/body mass (*top*) and leg power/body mass (*bottom*) in HR (●) and LR (○) to peak CK values.

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sity exceeds this permissible range, the membrane permeability temporarily changes, resulting in CK release from the active muscle. The boundary of this permissible range is its break point. In HR individuals, relatively greater muscle tension was required than in LR individuals to complete the same exercise. In addition, mobilization of FFAs, which acts as an energy substrate during exercise, tended to be lower in the HR than in the LR individuals. This suggests that the main active muscles of HR individuals might not be supplied with enough energy substrate to endure repeated tension during intense exercise. Therefore, the relative exercise intensity for the muscle seemed to rise according to developing muscle fatigue with continued exercise. It was estimated that metabolic enhancement of the glycolytic pathway induces the production of lactic acid. Thus the HR group seems to have exercised beyond this break point, resulting in an increase in serum CK activity.

In this study, we estimated levels of CK-MB. CK isozymes are classified into CK-BB, CK-MB, and CK-MM (5). CK-MB is expressed in the brain, smooth muscle, cardiac, and developing skeletal muscle and is released into circulation when myocardial infarction occurs. Some studies, however, have reported that circulating CK-MB levels increase after endurance exercise due to myocardial lesions (7) and damaged skeletal muscle (1). In this study, no significant changes were observed in serum CK-MB in both groups after exercise. This suggests that our endurance exercise test did not induce excessive muscle damage.

Thus this study demonstrates for the first time that there is a break point of CK release at 300–500 IU/l in serum after endurance exercise and that these values are associated with distinctive individual muscle properties. However, it must be recognized that this study is based on the use of just a single exercise loading and this break point may change for other exercise loadings. To establish the actual break point, there may be two approaches: use of the same or "absolute" workload or use of "relative" exercise loading. However, it is difficult to find an appropriate relative exercise loading that can take into account all physical characteristics. Thus we used the same absolute workload rather than relative exercise loading. If we want to realize our original aims, further studies that use alternative approaches (other exercise loadings) are required. If we can establish the threshold value, we can use this as an indicator to avoid overtraining in endurance exercise. It is recommended therefore to measure muscle strength for each individual before prescribing endurance exercise, because it is easier to measure muscle strength than to measure area or volume.

The authors thank the subjects for participation; Dr. Yoshinao Nakagawa, Otaru University, for assistance with the data analysis; and Drs. Yukio Sasaki and Hiraku Yodono of the Department of Radiology, Hirosaki University School of Medicine, for assistance in performing the magnetic resonance imaging.

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